

Exercise E3

We formatted the queries with the built in format button of this site:

<https://sparql.dblp.org/xmpyqZ>

a)

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT (COUNT(*) AS ?triples) WHERE {
  ?s ?p ?o .
}
```

b)

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT (COUNT(?publication) AS ?publications) WHERE {
  ?publication rdf:type dblp:Publication
}
```

c)

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT ?pers WHERE {
  ?pers dblp:orcid <https://orcid.org/0000-0002-5515-7158> .
}
```

d)

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT (COUNT(DISTINCT ?author) AS ?authorsWithWiki) WHERE {
  ?author dblp:wikipedia ?wikipediaPage .
}
```

e)

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT ?paper ?title WHERE {
  ?paper dblp:title ?title .
  FILTER (CONTAINS(?title,"CBR"))
}
LIMIT 10
```

f)

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT ?publishedIn (COUNT(DISTINCT ?publication) AS ?publications) WHERE {
  # Ralph and Alexander are identified via ORCID
  ?ralph dblp:orcid <https://orcid.org/0000-0002-5515-7158> .
  ?alexander dblp:orcid <https://orcid.org/0000-0002-2458-9943> .
  # Publications where both are authors
  ?jointPublication dblp:authoredBy ?ralph ;
```

```

        dblp:authoredBy ?alexander ;
        dblp:authoredBy ?coauthor .
# Exclude Ralph and Alexander
FILTER (?coauthor != ?ralph && ?coauthor != ?alexander)
# All publications of those co-authors
?publication dblp:authoredBy ?coauthor ;
        dblp:publishedIn ?publishedIn .
}
GROUP BY ?publishedIn
ORDER BY DESC(?publications)
LIMIT 10

```

```

g)
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
SELECT (COUNT(DISTINCT ?publication) AS ?publications) WHERE {
# Ralph and Ingo are identified via ORCID
?ralph dblp:orcid <https://orcid.org/0000-0002-5515-7158> .
?ingo dblp:orcid <https://orcid.org/0000-0002-3369-813X> .
?publication dblp:authoredBy ?author ;
        dblp:yearOfPublication ?year .
FILTER (?author = ?ralph || ?author = ?ingo)
FILTER (xsd:integer(STR(?year)) >= 2020)
}

```

```

h)
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?author (SAMPLE(?name) AS ?authorName) (SAMPLE(?affiliation) AS ?aff)
(COUNT(DISTINCT ?publication) AS ?publications) WHERE {
?author rdf:type dblp:Person ;
        dblp:creatorName ?name ;
        dblp:affiliation ?affiliation .
FILTER (CONTAINS(?affiliation,"Germany"))
?publication dblp:authoredBy ?author .
}
GROUP BY ?author
HAVING (COUNT(DISTINCT ?publication) >= 800)
ORDER BY DESC(?publications)
LIMIT 10

```

```

i)
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT ?coauthorName ?coauthorAffiliation (COALESCE(?sampleOrcid, "unknown") AS
?coauthorOrcid) ?firstAuthorPublications WHERE {
{

```

```

SELECT ?coauthor (SAMPLE(?name) AS ?coauthorName) (SAMPLE(?affiliation) AS
?coauthorAffiliation) (SAMPLE(STR(?orcid)) AS ?sampleOrcid) (COUNT(DISTINCT
?publication) AS ?firstAuthorPublications) WHERE {
  {
    # Find all co-authors of Ralph excluding himself
    SELECT DISTINCT ?coauthor WHERE {
      ?ralph dblp:orcid <https://orcid.org/0000-0002-5515-7158> .
      ?jointPublication dblp:authoredBy ?ralph ;
        dblp:authoredBy ?coauthor .
      FILTER (?coauthor != ?ralph)
    }
  }
  ?coauthor dblp:creatorName ?name .
  # Retrieve affiliation and ORCID if it exists
  OPTIONAL {
    ?coauthor dblp:primaryAffiliation ?affiliation .
  }
  OPTIONAL {
    ?coauthor dblp:orcid ?orcid .
  }
  # Select signatures where the co-author is the first author
  ?publication dblp:hasSignature ?signature .
  ?signature dblp:signatureCreator ?coauthor ;
    dblp:signatureOrdinal 1 .
}
GROUP BY ?coauthor
}
}
ORDER BY DESC(?firstAuthorPublications)
LIMIT 12

```

```

j)
PREFIX dblp: <https://dblp.org/rdf/schema#>
SELECT ?publication ?title ?firstAuthorName ?numberOfAuthors ?year WHERE {
  ?ralph dblp:orcid <https://orcid.org/0000-0002-5515-7158> .
  ?publication dblp:authoredBy ?ralph ;
    dblp:title ?title ;
    dblp:yearOfPublication ?year ;
    dblp:numberOfCreators ?numberOfAuthors ;
    dblp:hasSignature ?signature .
  ?signature dblp:signatureOrdinal 1 ;
    dblp:signatureCreator ?firstAuthor .
  ?firstAuthor dblp:creatorName ?firstAuthorName .
}
ORDER BY DESC(?year)
LIMIT 1

```

Screenshots of the results:

a)

Query results:

1

2,220ms

2,219ms

1ms

?triples

1

1,577,796,732

20th June 2026

b)

Query results:

1

19ms

18ms

1ms

?publications

1

8,597,634

20th June 2026

c)

Query results:

1

3ms

2ms

1ms

?pers

1

RalphBergmann

20th June 2026

d)

Query results:		1	2ms	2ms	0.0ms
		?authorsWithWiki			
1		9,901			

20th June 2026

e)

Query results:

10

66ms

65ms

1ms

	?paper	?title
1	 TreDCG08	Flexible Querying Techniques Based on CBR.
2	 ArcosML11	Using Introspective Reasoning to Improve CBR Sys[...]
3	 CheethamCG01	Applications of Soft CBR at General Electric.
4	 Diaz-AgudoG07	An Ontological Approach to Develop Knowledge Int[...]
5	 PietroGPS19	J2CBROKER as a Service: A Service Broker Simulat[...]
6	 KrishnanMSK18	Experiment: Investigating Feasibility of Coexist[...]
7	 MalandraSACH22	Community CBRS Networks - What You Need to Know.
8	 BahlsR07	Explanation Support for the Case-Based Reasoning[...]
9	 Hammond92	Reasoning as Remembering: The Theory and Practic[...]
10	 OuWLC05	Interactive Knowledge Validation and Query Refin[...]

20th June 2026

f)

Query results:

 10  33ms  32ms  1ms

	?publishedIn	 ?publications
1	CoRR	96
2	ICCBR	25
3	ETFA	22
4	GI-Jahrestagung	18
5	EnviroInfo	14
6	Business Process Management Workshops	13
7	ICPM Workshops	10
8	INFORMATIK	8
9	Softw. Syst. Model.	8
10	LWDA	8

20th June 2026

g)

Query results:

 1  23ms  22ms  1ms

	 ?publications
1	127

20th June 2026

h)

Query results:

10

317ms

316ms

1ms

	?author	?authorName	?aff	?publications
1	NassirNavab	Nassir Navab	Technical University Munich, Germany	1,784
2	2265	Robert Schober	Friedrich-Alexander University Erlangen-Nürnberg[...]	1,433
3	4934	Björn W. Schuller	University of Augsburg, Department of Computer S[...]	1,345
4	AloisKnoll	Alois C. Knoll	Technical University Munich, Department of Compu[...]	1,257
5	WilMPvanderAalst	Wil M. P. van der Aalst	RWTH Aachen University, Chair of Process and Dat[...]	1,252
6	7133	Andreas K. Maier	University of Erlangen-Nuremberg, Department of [...]	1,223
7	2478	Daniel Rückert	Technical University of Munich, Germany	1,169
8	2492	Giuseppe Caire	Technical University of Berlin, Department of Te[...]	1,132
9	GFettweis	Gerhard P. Fettweis	Aachen University of Technology, Germany	1,088
10	119	Bernhard Schölkopf	Max Planck Institute for Intelligent Systems, Tü[...]	1,069

20th June 2026

i)

Query results:

12 lines found 83ms in total 82ms for computation 1ms for resolving and sending

	?coauthorName	?coauthorAffiliation	?coauthorOrcid	?firstAuthorPublications
1	Yolanda Gil	University of Southern California, Information S[...]	https://orcid.org/0000-0001-8465-8341	89
2	Barry Smyth	University College Dublin, Insight Centre for Da[...]	https://orcid.org/0000-0003-0962-3362	87
3	Ulrich Furbach	University of Koblenz and Landau, Germany	https://orcid.org/0000-0002-3158-4446	74
4	Klaus Schmid	University of Hildesheim, Department of Computer[...]	https://orcid.org/0000-0002-4147-3942	69
5	Stefanie Rinderle-Ma	Technical University of Munich, Germany	https://orcid.org/0000-0001-5656-6108	56
6	Manuela Veloso	Carnegie Mellon University, Pittsburgh, USA	https://orcid.org/0000-0001-6738-238X	53
7	Karl Branting	-	https://orcid.org/0000-0002-9362-495X	53
8	Mirjam Minor	Goethe University of Frankfurt, Germany	https://orcid.org/0000-0002-6592-631X	52
9	Manfred Reichert	University of Ulm, Germany	https://orcid.org/0000-0003-2536-4153	49
10	Erica Melis	German Research Institute for Artificial Intelli[...]	unknown	47
11	Barbara Weber	University of St. Gallen, Switzerland	https://orcid.org/0000-0002-6004-4860	46
12	Klaus-Dieter Althoff	University of Hildesheim, Germany	https://orcid.org/0000-0002-7330-6540	45

20th June 2026

j)

Query results:

1 79ms 79ms 0.0ms

	?publication	?title	?firstAuthorName	?numberOfAuthors	?year
1	LenzDSB25	Numbers Don't Lie: Hybrid Extraction and Validat[...]	Mirko Lenz	4	2026

20th June 2026